L3.4 In-class Assignment – Looping

CPSC 292 – 10/28/2026

Name: _____ **Section:** _____

Collaboration team: _____

Question 1

Write a `for` loop that prints out each color of the rainbow for each loop in order (hint: ROY G BIV).

Question 2

Write a `repeat` loop that breaks after 1,000 steps. Have the loop print out the step.

Question 3

For the vector `my_dat`:

```
my_dat <- c(8, NA, 10, 9, 2, 1, 7, 0, NA, 5)
```

write a loop that cycles through each position of the vector and either prints the value if it is a number or skips if the value is `NA`. (Hint: use `next` to skip.)

Question 4

The factorial operator "x!" multiples each number from x to 1 by itself. So, 3! = 3 * 2 * 1 = 6. It can be expressed mathematically as:

$x! = x \times (x-1) \times (x-2) \times ... \times 1$

Use a while loop to calculate the factorial of a number and stores the factorial in a new object. There is a special case for factorial where 0! = 1, make sure your code will return this value correctly.

Question 5

Use `apply()` to calculate the mean of the numbers in `mat1`:

a. by row.
b. by column.

```
> mat1
           [,1]         [,2]       [,3]       [,4]       [,5]
 [1,] 0.8737614 3.279995e-01 0.99382677 0.04708227 0.66861009
 [2,] 0.1350519 8.806709e-01 0.19290140 0.58394507 0.91417062
 [3,] 0.3590008 4.981267e-01 0.16924711 0.70743048 0.93737118
 [4,] 0.8016250 2.561859e-05 0.60500166 0.97617265 0.02148424
 [5,] 0.4270226 9.278873e-01 0.97623484 0.53537165 0.62143089
 [6,] 0.8942769 1.397935e-01 0.25193916 0.35950828 0.34564422
 [7,] 0.4148868 1.625391e-01 0.36599624 0.53433252 0.84628688
 [8,] 0.3866375 7.406660e-02 0.04243032 0.37379779 0.55480836
 [9,] 0.6609133 7.499941e-01 0.39202963 0.26426009 0.91887630
[10,] 0.2191938 5.467871e-02 0.91405195 0.74919128 0.55894999
```

Figure 1: mat1